

BIOSTEC 2026

19TH INTERNATIONAL JOINT CONFERENCE ON BIOMEDICAL
ENGINEERING SYSTEMS AND TECHNOLOGIES

2 - 4 MARCH, 2026

Marbella, Spain

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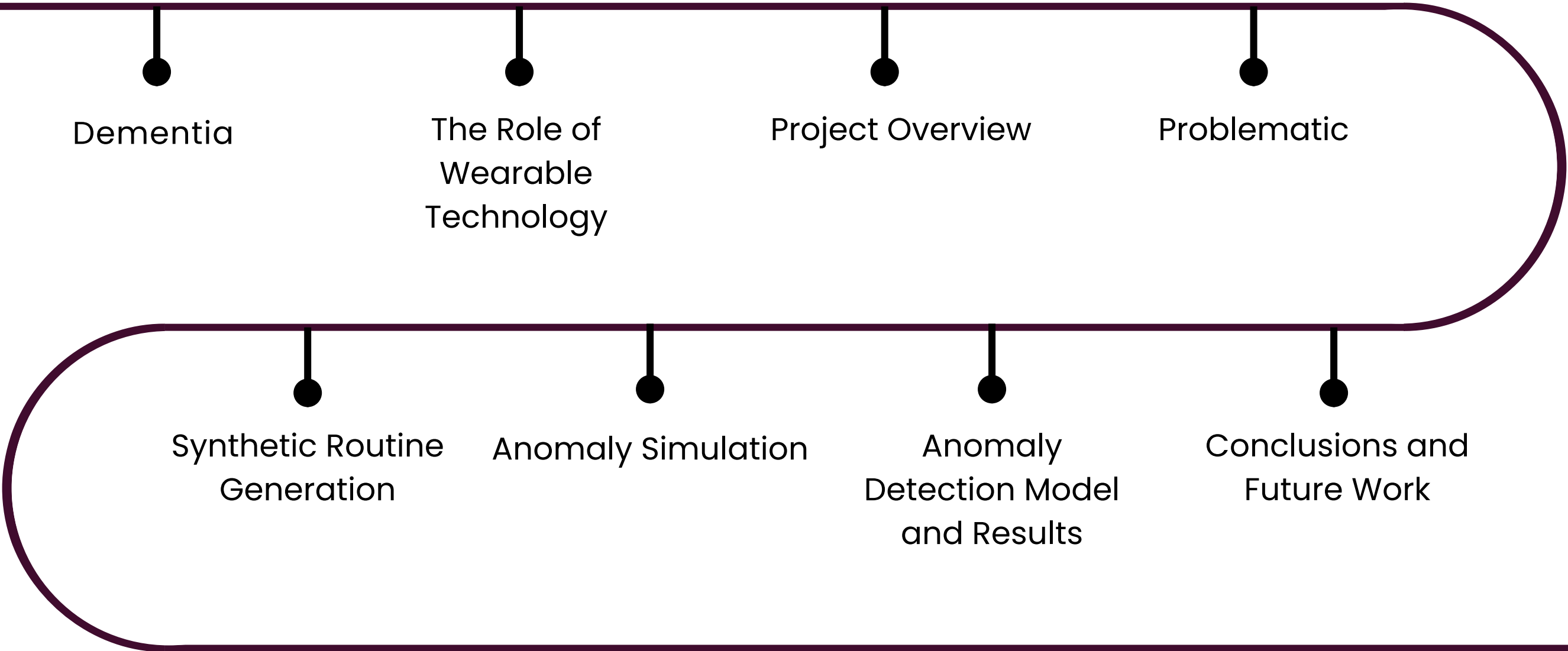
Towards Synthetic Generation of Daily Routines and Anomaly Detection with Wearable Sensors for Dementia Care

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Dementia

There are more than 10 million new cases of dementia each year worldwide, which implies a **new case every 3.2 seconds**.

Among the most significant challenges in dementia care is **gradual loss of autonomy** and disruption of daily routines, which can severely **impact both patient well-being and caregiver burden**.



Tracking movement, sleep, and health indicators improves safety and independence



The Role of Wearable Technology



CONTINUOUS MONITORING

Wearables track movement, sleep, and vital signs in real time



ANOMALY DETECTION

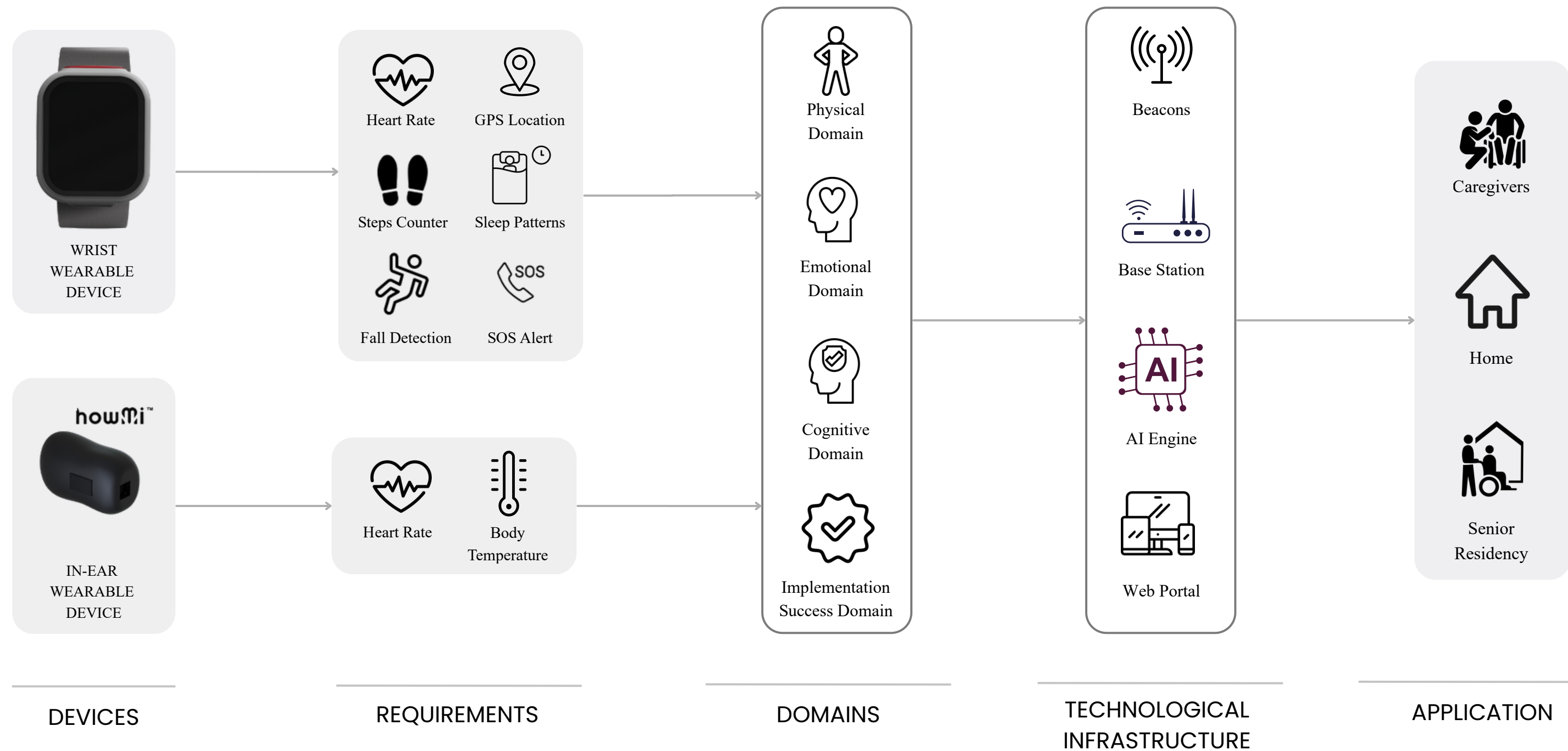
AI-powered wearables identify abnormal behaviors and alert caregivers



IMPROVED QUALITY OF LIFE

Supports independence and reduces caregiver burden through early intervention

Project Overview



Problematic

Limited real-world dementia data

Most wearable datasets are collected from **healthy adults**

Data from people with dementia is **scarce, sensitive, and hard to annotate**

Beyond activity recognition

Recognising activities is not enough

Detecting deviations from daily routines is essential for early intervention in dementia care

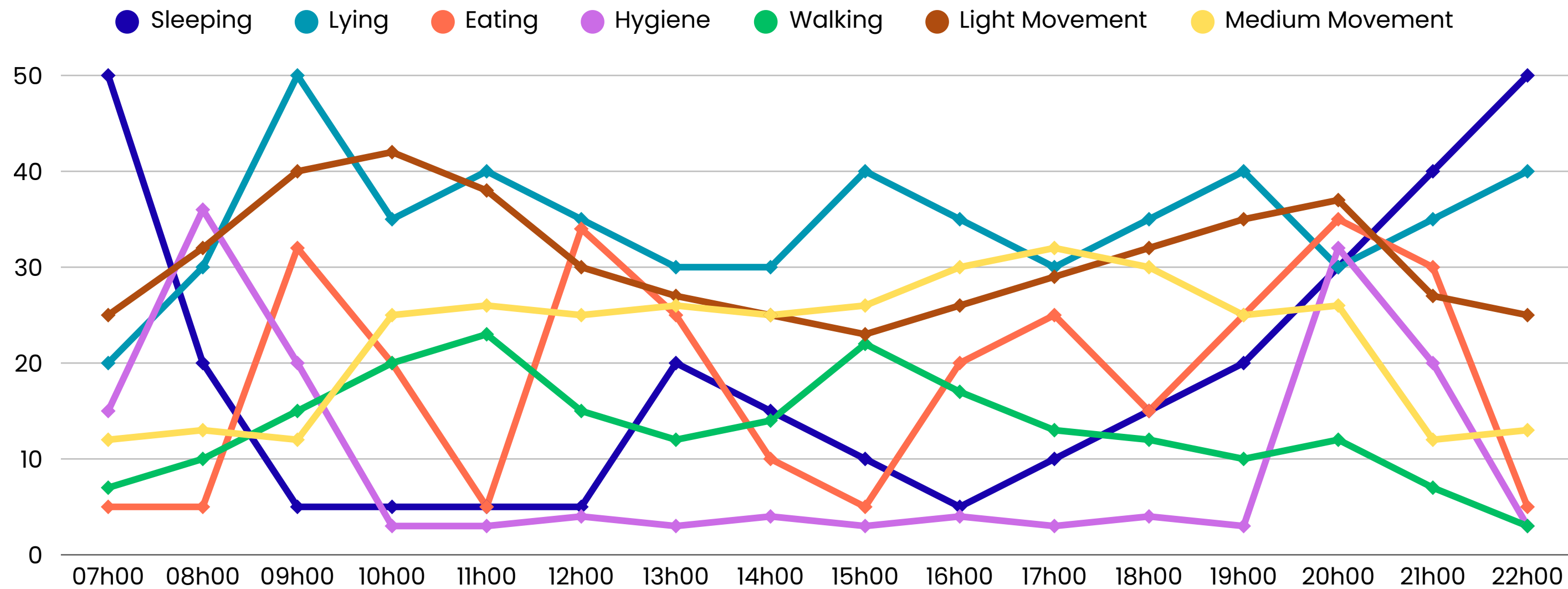
Complex nature of behavioural anomalies

Anomalies are context-dependent, subtle, and highly variable

High risk of false positives in real-world deployments

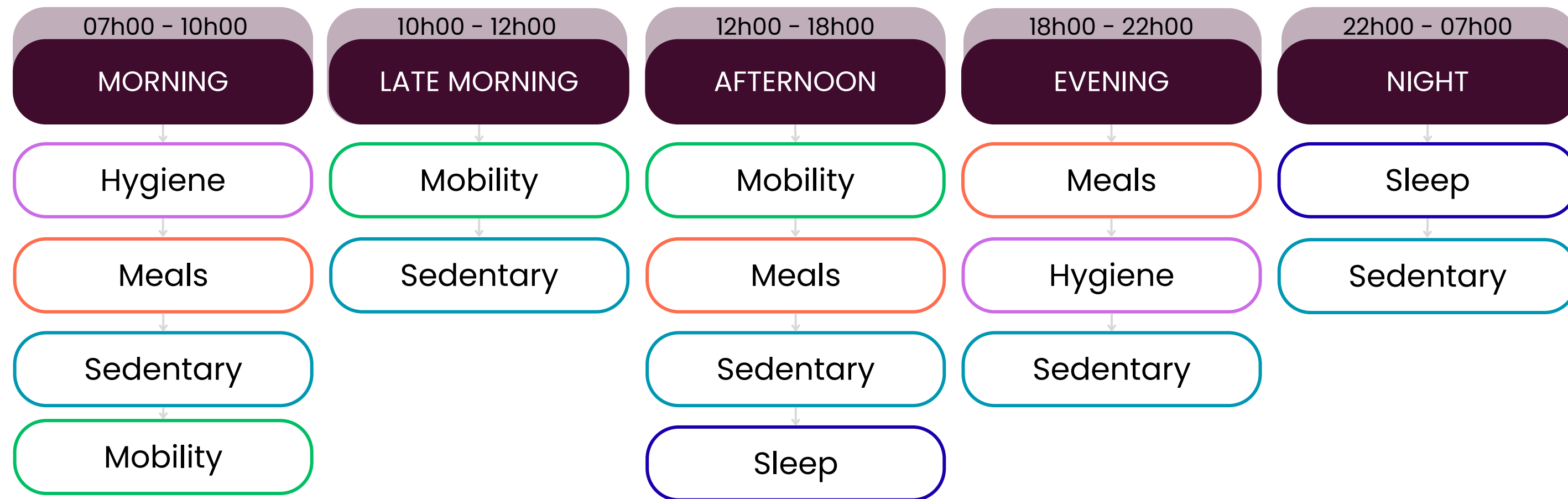
Synthetic Routine Generation

The distribution of activity categories throughout the day



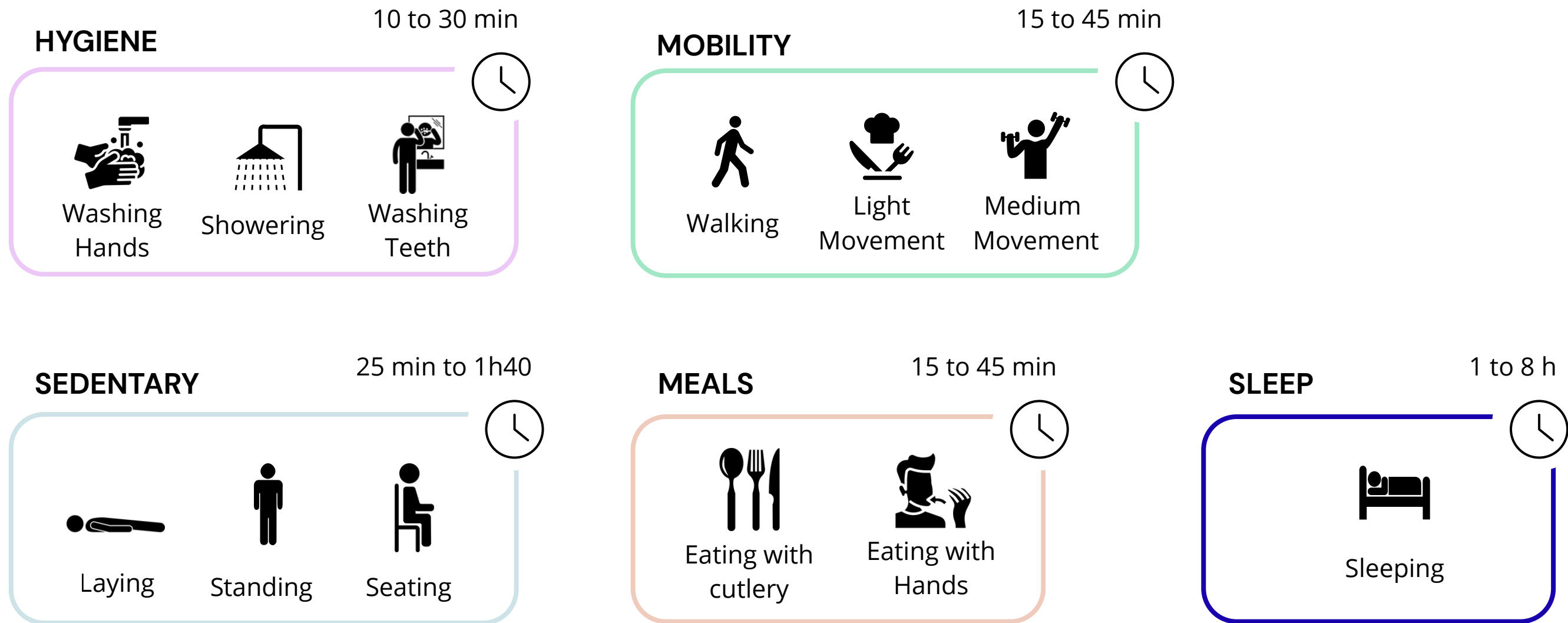
Synthetic Routine Generation

The typical daily flow used to generate these routines

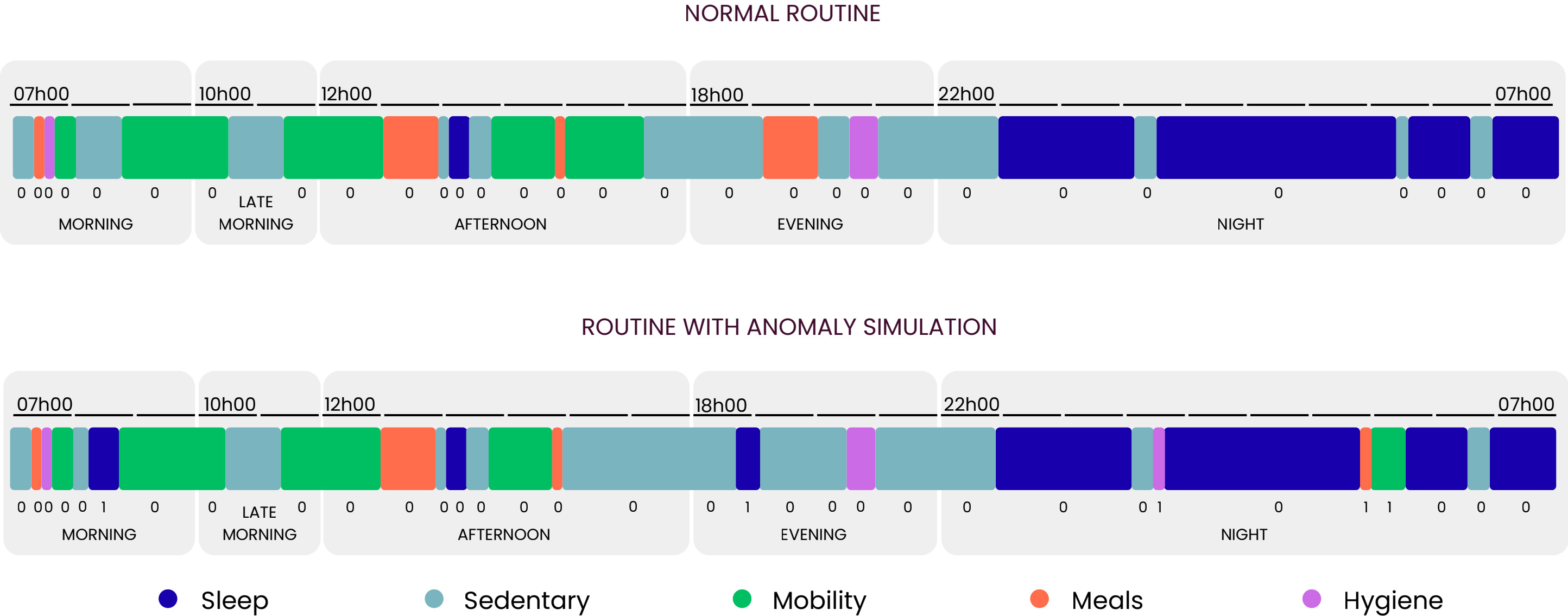


Synthetic Routine Generation

Activity Duration

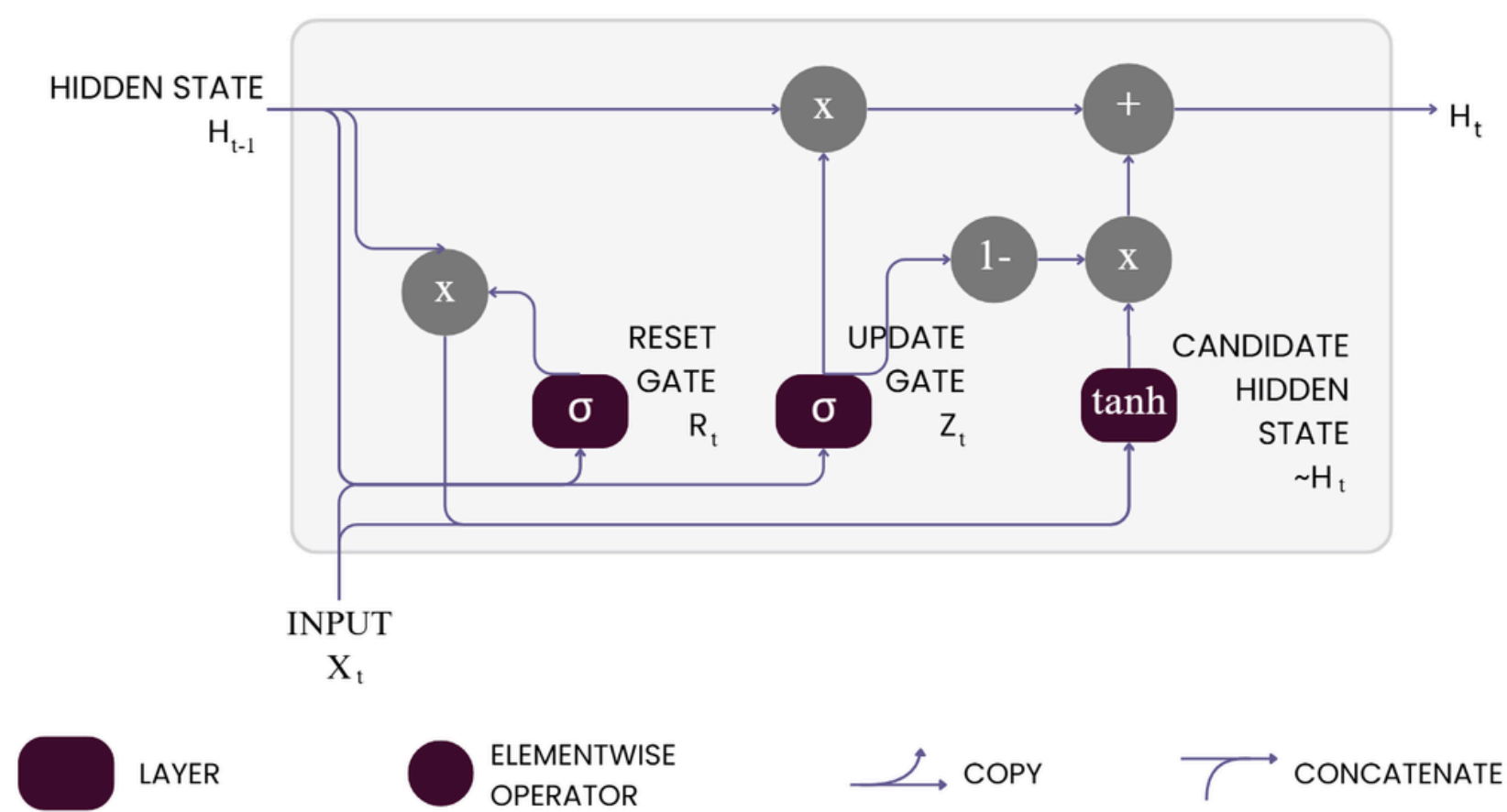


Anomaly Simulation



Anomaly Detection Model and Results

GRU Model



Results

	Precision	Recall	F1-score
Normal	0.995	0.998	0.997
Anomalous	0.978	0.941	0.959

AUROC: 0.999
PR-AUC: 0.989

Conclusions and Future Work

Effective learning of daily routine structure;
Strong anomaly detection performance (AUROC, PR-AUC);
Errors mainly in temporally flexible activities.



Validation with real wearable data and adaptation for real-time deployment

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THANK YOU!

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